Freight Cost per Kilogram — 2023 vs 2026 (Black King / Coopercabruca lanes)

Prepared by Sophia (TrueSight DAO Autopilot) · 2026-10-13 · **Analysis only — no payment made or implied.**

1. The two shipments

	2023 shipment	2026 shipment (AWB 047-3175-3223)
Origin lane	Coopercabruca → Salvador (SSA)	Black King, Ilheus → Salvador (SSA)
Flight	TAP 047-04136716	TAP 047-3175-3223
Cargo	100 kg cacao nibs	9 articles cacao (beans/nibs/mass/tea/butter/almonds)
Net / gross	100 kg / 104 kg	302.06 kg / 349 kg
Boxes / pallets	8 volumes	27 boxes · 2 pallets
Forwarder	Seacoast Logistics (US-side only)	5 Continent #1026-08 (end-to-end)
Status	Delivered	**GATED — awaiting physical arrival**

2. Cost side-by-side (all known cost, USD)

Cost bucket	2023	2026
Forwarder invoice (as billed)	521.03	4,866.33
Brazil-side, billed separately (confirmed)	943.89	136.69
Brazil-side, billed separately (possible, unconfirmed)	0.00	261.75
TOTAL KNOWN	**1,464.92**	**5,003.02 → 5,264.77**

Breakdown of the "separately billed" buckets:

Separately-billed line	2023	2026
Omega numerario — air freight SSA→SFO	776.20 (R$4,000)	—
CT-e inland transport, coop→Salvador	167.69 (R$864.15)	—
Cintagem (re-strap, 02/10)	—	58.17 (R$300)
Fumigation NFS-e (ASTRA, 24/09) — include only if export-prep	—	78.53 (R$405)

Separately-billed line	2023	2026
Diaria / truck demurrage — POSSIBLE, not confirmed	—	261.75 (R$1,350)
Airline reissue fee	—	unquantified

3. Per-kilogram — the headline comparison

Basis	2023	2026 (invoice only)	2026 (+ confirmed extras)	2026 (+ diaria)
Per gross / chargeable kg	**$14.09** (104 kg)	$13.94 (349 kg)	**$14.34**	$15.08
Per net cargo kg	$14.65 (100 kg)	$16.11 (302.06 kg)	$16.56	$17.43

4. What the numbers say

- **All-in per kg, the two years are nearly identical** — ~$14.09 vs ~$14.34 per gross kg. The "2026 invoice is 9.3x the 2023 invoice" headline does NOT translate into a per-kg blowout.
- Most "new" 2026 invoice lines are work that in 2023 sat on a **separate Brazil invoice** (inland transport, air freight) or are **weight-scaled** (air freight, duties: 349 vs 104 kg = 3.4x).
- Genuinely structural adds (palletizing, per-line fee, duty-advance, bank fee) are small once spread over 349 kg.
- **2026 will drift UP, not down:** (a) $4,866.33 is superseded — a US-bond line is being added back; (b) $414.59 export-customs may be a DUPLICATE if Omega bills separately (exactly the 2023 pattern); (c) diaria and reissue fee are unquantified.

5. Benchmark — DHL / FedEx express vs the forwarder

Transport basis	Weight band	Indicative USD/kg	Source / basis
Forwarder air-freight line, 2026 actual	349 kg	~3.56	5 Continent invoice: $1,241 / 349 kg
Forwarder air-freight line, 2023 actual	100 kg	~7.76	Omega numerario: R$4,000
Air-freight market, Brazil→USA	~200 kg	2.50–4.50	Suaid Global, Sep 2026
Air freight all-in (fuel, security, handling, customs)	any	4.00–10.00	ExFreight, 2026
Express-courier premium over standard air	any	+20–40%	ExFreight, 2026

Transport basis	Weight band	Indicative USD/kg	Source / basis
DHL Express Worldwide heavy-weight, published retail	68–150 kg (150–330 lb)	18.70–84.80	DHL US 2026 Rate Guide, per-lb multiplier x 2.2046
DHL Heavy Weight Express service	50–3,000 kg	quote-only, palletized	DHL Brazil 2026 Rate & Service Guide
FedEx vs DHL, same international route	small parcel	FedEx typically 20–40% higher	Easyship, 2026

Read:

- Express couriers are priced **per parcel / per-kg on a different cost curve** than freight forwarding. For a 100–350 kg palletized cacao export, express retail lands **roughly $15–40+/kg** for the transport leg — versus the **~$3.50/kg** the DAO pays on the forwarder's air-freight line, and **~$14/kg all-in** including customs, FDA, duties, inland, and delivery.
- So the DAO's forwarder lane is **~4x–10x cheaper on the transport line alone** than DHL/FedEx express retail for this weight class. Express is viable only for urgent, small, high-value parcels — not bulk cacao.
- Caveat: published express rates are **list/retail**. Business accounts get up to ~40% off (DHL 2026), and heavy-weight tiers are quote-only. A real DHL/FedEx quote would need exact dims + origin/destination — recommend requesting one if a formal comparison is wanted.

6. Prior DAO research (recovered)

Your earlier freight-costing work lives in a Google Sheet, not in the context repo:

- **Sheet:** "Costing estimates" — id `10Ps8BYcTa3sIqtoLwlQ13upuxIG_DgJIpfzchLjm9og`
- **SA:** `agroverse-market-research@get-data-io.iam.gserviceaccount.com`
- **Structure:** Quick Estimate dashboard · Cost Breakdown (200/300/500/750/1000 kg tiers) · Totals by Weight · Notes & Assumptions
- **SeaCoast air-freight tiered rates (USD/kg):** 200 kg → 3.50 · 300 kg → 3.40 · 500 kg → 3.30 · 750 kg → 3.30 · 1000 kg → 3.20
- **Script:** `go_to_market/airfreight_costing/parse_seacoast_quotation.py` (SPREADSHEET_ID hard-coded)
- **Repo:** `TrueSightDAO/agroverse-freight-audit` — `pointers/freight_lanes.json` (machine tiered-per-kg + lump-sum lanes) and `snapshots/` (dated vendor quotes).

7. Caveats

- **FX wobble (2023):** the BRL legs are converted at the DAO's standing ledger rate 1 BRL = 0.19405 USD, not the Nov-2023 spot (~0.205). At spot, 2023 Brazil-side ≈ $997 → total ≈ $1,518 → ~$14.60/kg gross. True 2023 USD likely sits between the two.
- **2026 FX** uses official BACEN PTAX 5.1575 (18/09/2026).

- **Goods value excluded on both sides** — this is FREIGHT / LOGISTICS only, not landed cost. 2026 goods = $6,473.02 (Rev 15). 2023 goods are not ledger-documented.
- **2026 cargo is under the arrival gate** — nothing here touches balances or lands inventory.